

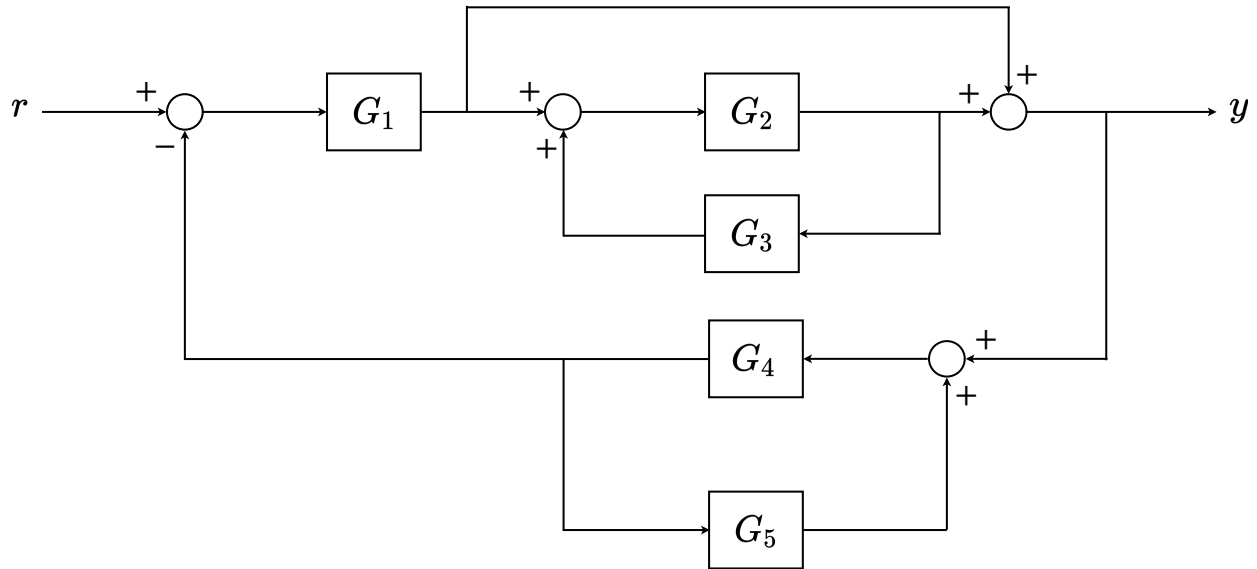
F26 AerE 3310: Flight Control Systems I

Practice Problem Set 4: Block Diagram: Reductions and Compositions

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Problem 1



Derive the overall SISO transfer function from r to y for the above interconnection. Your final answer must be in terms of the SISO transfer functions G_1, G_2, G_3, G_4, G_5 shown. Show all your intermediate figures and/or calculations for this derivation.

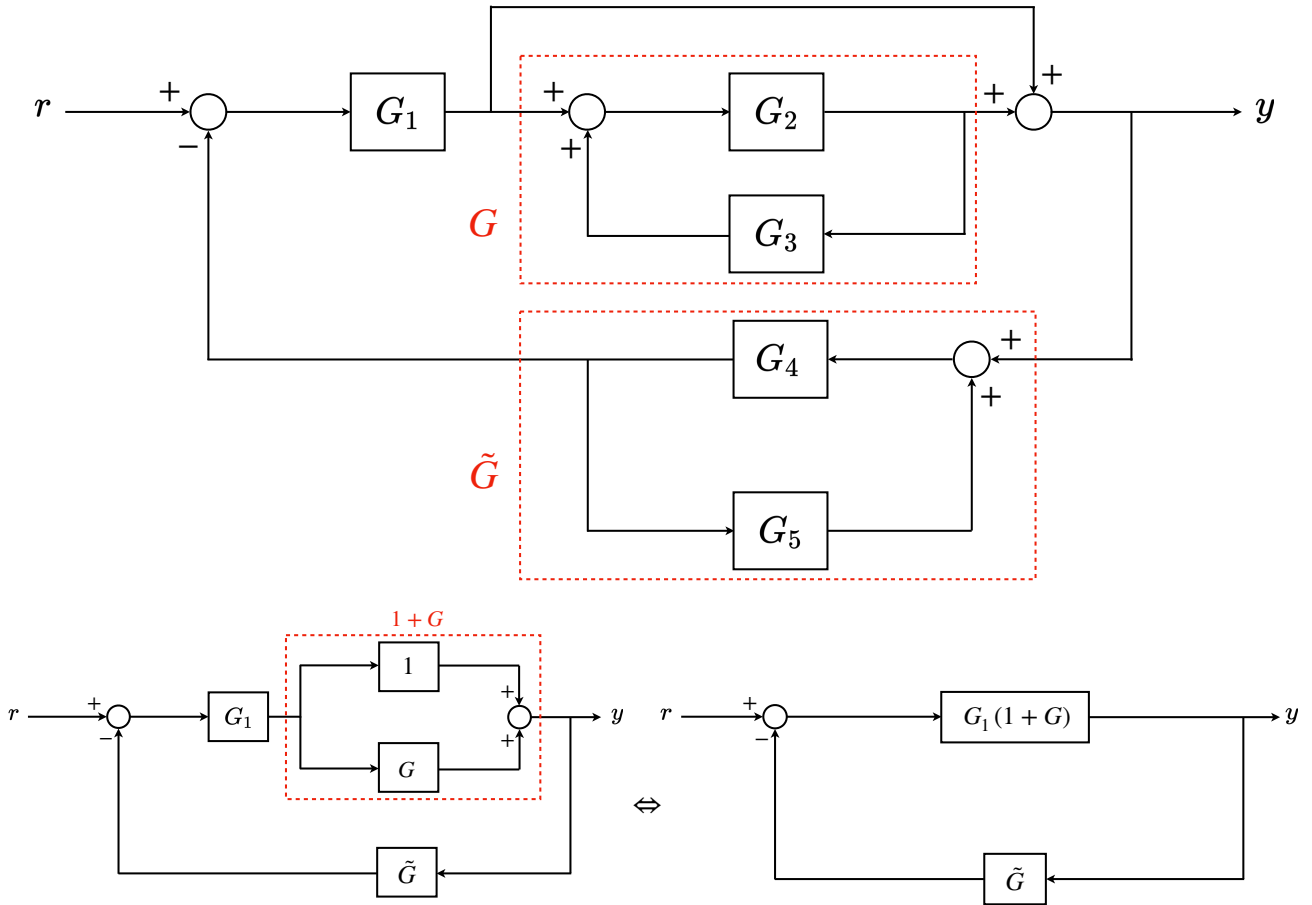
Solution for Problem 1

We first deduce the transfer functions $G, \tilde{G}$ for the dashed boxes shown in the figure above, both of which are positive feedback interconnections. So by Lec. 11 p. 6, we have

$$G = \frac{G_2}{1 - G_2 G_3}, \quad \tilde{G} = \frac{G_4}{1 - G_4 G_5}.$$

Using Lec. 11 p. 3, we then reduce parallel connections of 1 and G giving $1 + G$, followed by series connection (Lec. 11 p. 2) of G_1 and $1 + G$, giving $G_1(1 + G)$, as shown below.

Finally, using Lec. 11 p. 4-5, the negative feedback interconnection of $G_1(1 + G)$ and $\tilde{G}$ gives



the overall transfer function

$$\begin{aligned}
 & \frac{G_1(1+G)}{1+G_1(1+G)\tilde{G}} \\
 &= \frac{G_1 \left(1 + \frac{G_2}{1-G_2G_3} \right)}{1+G_1 \left(1 + \frac{G_2}{1-G_2G_3} \right) \left(\frac{G_4}{1-G_4G_5} \right)} \\
 &= \frac{G_1(1+G_2-G_2G_3)(1-G_4G_5)}{(1-G_2G_3)(1-G_4G_5)+G_1(1+G_2-G_2G_3)G_4}.
 \end{aligned}$$

Problem 2

Consider the following two transfer functions:

$$G_1(s) = \frac{3}{s+2}, \quad G_2(s) = \frac{5}{s-1}.$$

In MATLAB documentation, look up the inbuilt commands `series`, `parallel` and `feedback` to compute the composed transfer functions for series, parallel and negative feedback interconnections in code, instead of computing by hand as we did in Lec. 11.

Write a MATLAB code that computes the composed transfer functions G_{Series} , G_{Parallel} , $G_{\text{NegFeedback}}$ for the given G_1, G_2 using the above inbuilt commands. Then use the same code to plot the unit step responses for G_{Series} , G_{Parallel} , $G_{\text{NegFeedback}}$ for $t \in [0, 2]$ in a single figure, as we did in Lecture 10 MATLAB demo: CANVAS file: `Lecture10MATLABDemo.m`.

Argue why your MATLAB plot makes sense by computing the poles (either by hand, or by MATLAB command `pole`) of the composed transfer functions.

Solution for Problem 2

Please find the MATLAB code `PS04p2.m` posted in CANVAS Files section, folder: Practice Problems and Solutions.